A Comparative Study of Plant Disease Detection Using Machine Learning

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Abstract*—Plant diseases are one of the major factors that can reduce crop quality and agricultural productivity worldwide. It can easily damage our food supply. Early detection of plant disease is very much needed for reducing damage and improving sustainable agricultural productivity. Traditionally plant diseases are identified by human experts through visual inspection of plant leaves, which can be time consuming and sometimes inaccurate. With the rapid development of machine learning and computer vision techniques we can classify and identify plant disease at a very early period of time. Images of plant leaves are used as input to train different machine learning models to identify whether a plant is healthy or affected by a disease.Before training the models, the images are preprocessed to improve their quality and make them suitable for analysis.The dataset consists of labeled images of healthy and diseased plant leaves which are collected from publicly available agricultural datasets. In this study, we will use several machine learning models such as AdaBoost, Random Forest, K-Nearest Neighbors (KNN), Convolutional Neural Networks (CNN) and Transfer learning Models etc are evaluated and implemented. Before training the models, the images are preprocessed improving their quality and making them suitable for analysis. The models are tested using a dataset of plant leaf images, and their results are measured using different performance metrics. Some image processing techniques such as normalization, resizing, noise reduction and feature extraction are applied to enhance the quality of the input data.The performance of the selected algorithms is compared using evaluation metrics including precision, recall and F1-score etc.The experimental results demonstrate the effectiveness of image based machine learning approaches in identifying plant diseases with deep learning models showing superior performance in classification accuracy. The findings of this research contribute to the development of intelligent agricultural systems that can assist farmers and agricultural experts to detect the plant disease through automated image analysis.

***Index Terms*—Random Forest, Adaptive Boosting(Adaboost), K-Nearest Neigbours (KNN), Convolutional Neural Networks (CNNs), Transfer Learning Models.**

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I. Introduction

Agriculture plays a vital role in many economies of many countries and it is essential for ensuring global food security. Plants are the primary source of living organisms for humans, animals. Healthy crops are necessary for growing demands of the world population. For this reason, plant diseases remain one of the major threats to agricultural productivity.These diseases can significantly affect food quality and can reduce crop yield along with substantial economic losses for the agricultural industries.That is why for maintaining healthy crops and improving agricultural productivity, early detection and accurate identification of plant diseases are very much needed.Our aim is to efficiently and accurately detect the plant pathogens and other factors in the early stage of the plants, which will become the cause of gradually decrement of the economic losses based on agriculture and getting quality full food intake, these will become the factor of safeguard for millions of lives across the world and their sound health.

Plant diseases are caused by various pathogens such as: bacteria, fungi and parasites. There are some common symptoms including leaf spots, wilting, mold growth, discoloration and abnormal patterns on plant surfaces. If these diseases are not detected at an early stage, they can spread rapidly and severely damage crops and reduce our economic growth. For this reason farmers have to monitor their crops to identify signs of diseases and apply appropriate treatments. Traditionally, the plant disease detection method has been performed manually from the farmers. Most of the time, many farmers do not have sufficient expertise to distinguish between similar symptoms caused by different pathogens and different diseases. On the other hand, manual inspection is very labor-intensive and time-consuming, especially in large agricultural fields. In today's world, with the advancement of technology, modern agricultural systems are increasingly adopting to automate solutions to address these problems. Artificial Intelligence, Machine Learning and

Computer vision have emerged as powerful tools for plant disease detection. These Artificial intelligence technologies can assist computers to analyze images of plant leaves so that farmers can identify disease symptoms with high accuracy.

To sum up, we can say that plant disease detection is needed in reducing economic losses, increasing food quality and crop yielding percentage. If we can detect the disease easily before using digital equipment like AI, we can further imply the quality of full seeds in the field and take precautionary measures earlier. While plant diseases are a great threat to the economic stability, particularly in heavily agricultural dependent countries such as Bangladesh- the manual inspection methods are too slow and labour-intensive. To highlight the critical issues, our project proposes an digital automated and early detection process using Artificial Intelligence and Machine Learning, image processing.

II. Literature Review

Over time, plant disease detection research has evolved from traditional image processing techniques to advanced machine learning and deep learning approaches. According to Balafas et al. [1], plant disease detection has shifted from traditional machine learning (including SVM and Random Forest) to automated deep learning. Convolutional Neural Networks (CNNs) perform accurately for both classification and object detection, although high computational costs and the lack of real-world datasets remains a challenge. They further added CNNs and object detection models such as YOLO dominate modern research, but their real-world application is hindered by background and lighting variations. There is a need for lightweight models for practical agricultural environments.According to Khalid and Karan [2], DL models offer scalable solutions, including strong architectures such as MobileNet with 96% accuracy and CNNs with 89% accuracy. Moreover, explainable AI like 'Grad-CAM' is effective for visualizing the classification and improving trustworthiness, though dependency on massive datasets persists.

Ahamed et al. [3], said that to highlight the need for an automated field solution, researchers developed a lightweight PDSCNN integrated with a RELM classifier, which contains only 9 layers and 0.053 million parameters. Environmental variations still cause some misclassifications, even after PDSCNN was deployed on Jetson Nano hardware, achieving 94.22% accuracy in 91 diseases. Gajjar et al. [4] also proposed a real time disease detection system deployed on a Nvidia Jetson TX1,integrating a Single Shot Detector(SSD) with the MobileNet V1 backbone for multi-leaf localization, and a custom CNN inspired by CNN containing approximately 6 million parameters for subsequent disease classification. CNN achieved 96.88% accuracy across multiple crops with an inference time of approximately 3 milliseconds per frame.Unfortunately, this study misclassified between visually similar diseases .

In another approach, Sakkarvarthi et al. [5] focused on targeting tomato crop diseases; a CNN model was built to perform complex architectures, including VGG19 and ResNet152, having 88.17 % testing accuracy in the PlantVillage dataset. The focus should be on optimizing this model and trying it for other crops in the future. In another study featuring tomato crops, Billah et al. [6] proposed a modified DenseNet21 architecture by reducing the number of convolutional layers to a light weight 14-layer network with 297,414 trainable parameters. Although the study lacked real world IoT deployment,the model achieved a testing accuracy of 98.62 % . In another approach, Pandian et al. [7] expanded a limited training dataset to more than 240,000 images using Generative Adversarial Networks (GAN) and Neural Style Transfer (NST). Although a wider variety of disease classes needs to be optimized for better generalization, the optimized five-layer DCNN achieved an accuracy of 98.41% .

Additionally, Vishnoi et al. [8] developed a low-resource CNN with 98 % accuracy for apple leaf diseases by addressing computational and storage limits. While this model is ideal for handheld devices, it requires testing on images from growth stages and diverse geographical locations to guarantee real-world robustness. Moreover, Afifi et al. [9] addressed the limitations of the requirement for large labeled dataset by evaluating several few-shot learning(FSL) strategies consisting of Transfer Learning baselines, a Triplet Network and Deep Adversarial Metric Learning(DAML) that use ResNet backbones. Their model classified novel diseases from as few as 5 to 50 images per class, achieving 99 % accuracy when the source and target domains were similar. Finally, Aboelenin et al. [10] proposed a hybrid deep learning framework combining an ensemble of three pre-trained CNNs: VGG16, Inception-V3, and DenseNet201- for global feature extraction with a Vision Transformer for fine-grained local features. Despite having a drawback for high computational cost of ViT, their models achieved an accuracy of 99.24 % on the Apple dataset and 98 % on the Corn dataset with close to perfect AUC-ROC scores

III. Methodology

A. Model Architecture

We followed some steps to build our plant disease detection project.The model architecture is illustrated below:

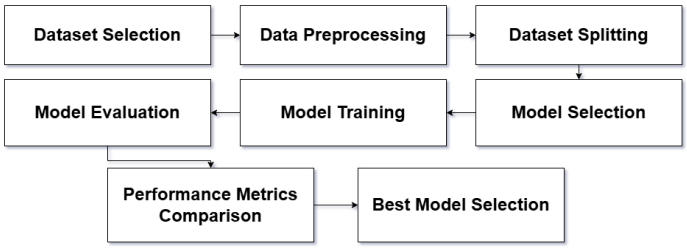


Fig. 1. Flowchart depicting the steps of methodology

B. Dataset Description

The dataset used in this study is the PlantVillage dataset, was collected from the open source platform Kaggle [11]. The PlantVillage dataset consists of 54305 images of healthy and diseased plants covering 14 species of crops. The dataset

contains labeled plant images across 38 distinct classes, covering a wide range of crops including tomato, corn, grape, potato, and others. In addition, it consists of 1 disease caused due to mites, 2 diseases caused by mold, 2 viral diseases, 2 bacterial diseases and 17 basic plant diseases. The images were arranged in class-specific folders and originally pre-divided into train and validation subsets. The dataset provides plant images under diverse environment conditions such as: varying lighting and background conditions, making it fit for multi-class image classification using machine learning and deep learning models.

C. Data Preprocessing

The dataset goes through a structured preprocessing pipeline to ensure there is consistency and quality across all model inputs. The original dataset, initially divided only into training and validation folders, was first merged into a single dataset by combining files class by class. This step was essential to remove any bias from the original arbitrary split and to enable a clean, controlled redistribution. The dataset was split into three subsets after merging all, with a fixed ratio of 70% for training, 15% for validation, and 15% for testing:

- Training set (70%) — used to train all models
- Validation set (15%) — used to monitor performance and detect overfitting during training
- Test set (15%) — reserved for final unbiased evaluation Each class folder was split independently to maintain class balance across all subsets. A fixed random seed (seed = 42) was used throughout to ensure reproducibility. Image augmentation was applied only to the training data for increasing the diversity and size of the training set. Three augmentation techniques were employed:
- Horizontal Flip - Flips each training image about its vertical axis to provide more spatial invariance
- Random Crop — a random region covering 85% of the image is cropped and resized back to original dimensions, simulating variation in framing

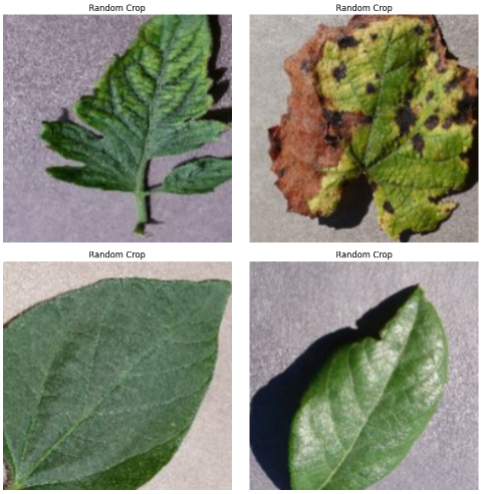


Fig. 2. Random Crop

- Gaussian Noise — added pixel-level noise to the normalized image, sampled from a Gaussian distribution with mean=0 and std=0.025 to improve robustness against sensor noise

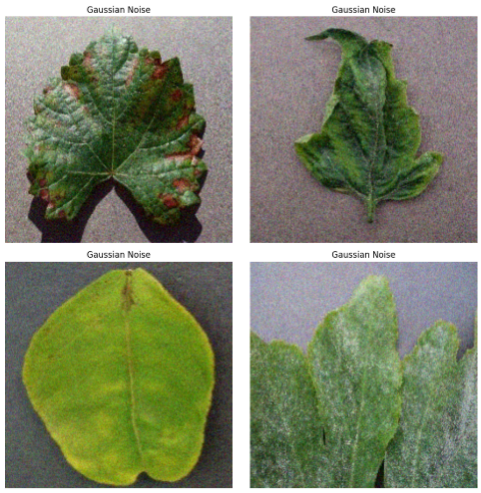


Fig. 3. Gaussian Noise

- Color Channel Flip — the three RGB channels are randomly permuted to prevent over-reliance on specific color patterns

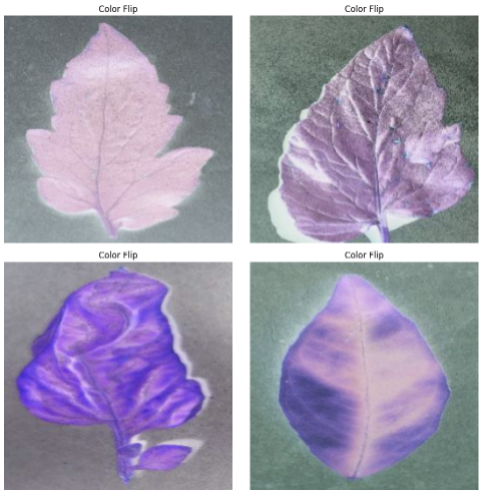


Fig. 4. Color Channel Flip

- Augmentation—- Each augmentation is applied with a probability of 50%, resulting in one augmented copy for each original training image. For a fair and unbiased evaluation across all models, no augmentation was applied to the validation and test sets. Images were loaded with resolution (224×224), and pixel values were normalized to [0, 1] for deep learning models.. For traditional ML models, images were resized to 64 x 64 pixels, flattened into 1D feature vectors, and then transformed via Principal Component Analysis to reduce its dimensionality to

100–200 components while keeping around 85–90% of the total variance.

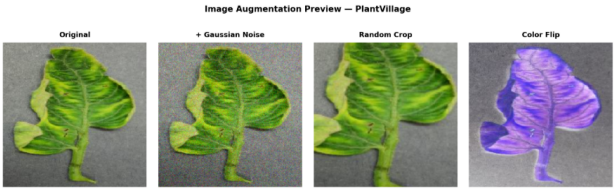


Fig. 5. Sample leaf image after augmentation

D. Learning Phase

Each of the model utilized the test data to predict and then evaluate the labels against the actual labels using standard classification metrics. Four machine learning models and a transfer learning deep learning-based model were trained on training, validation, and test data:

a) Convolutional Neural Network (CNN): CNNs are made to automatically use convolutional procedures to learn organizational structures of characteristics from raw picture pixels. Three convolutional blocks—a Conv2D layer, BatchNormalization, and MaxPooling2D—were used to create a bespoke sequential CNN. To reduce overfitting, which a GlobalAveragePooling2D layer replaces conventional flattening. The final softmax output comes after a completely linked Dense(128) layer with Dropout(0.5). The Adam optimizer was used to create the model, and sparse categorical cross-entropy loss was used to train it for ten epochs.

b) Random Forest (RF): Random forest is a machine learning model which combines several decision trees to reduce the correlation and improve performance among feature data [12]. A huge advantage of Random forest is that it functions effectively on large datasets and can hold many variables helping us to analyze large numbers of input variables without deleting them. Additionally, it overcomes the overfitting problem by aggregation of multiple decision trees which reduces bias and volatility. Even in the absence of hyperparameter adjustment, it yields superior outcomes. Each tree is trained on randomly selected input features and samples. A forest of 100 estimators was trained on PCA related features utilizing all CPU cores parallely for this model.

c) AdaBoost: Adaboost also referred to as Adaptive Boosting is an ensemble technique which combines multiple weak classifiers to build a strong model. It focuses more on the misclassified data points from previous models for generating a strong classifier [13]. We used a decision tree stump of max depth 1 as the base estimator with 50 boosting rounds and considering a learning rate of 1.00. Additionally, Adaboost reuses the same PCA reduced feature space we used in Random forest which leads to a consistent and fair comparison .

d) K-Nearest Neighbors (KNN): The majority vote of the k nearest neighbors in the feature space determines a class label for KNN, a non-parametric lazy learning method [4]. There is no need for an particular training period. On PCA-reduced features, k=5 neighbors with Euclidean distance are used for classification. KNN is a crucial benchmark for assessing the quality of the retrieved feature representations, despite its simplicity.

e) ResNet50 (Transfer Learning): Rich hierarchical visual information may be extracted using ResNet50, a 50-layer deep residual network that was pre-trained on ImageNet. The basic model was utilized just as a feature extractor and loaded with frozen weights (trainable = False). GlobalAveragePooling2D, Dense (256, relu), Dropout (0.5), and Dense (num_classes, softmax) were added to a custom classification head, which was trained for ten epochs with the Adam optimizer at a learning rate of 0.001. With little domain-specific training data, this method, known as feature extraction transfer learning, achieves great accuracy by using pre-learned representations.

IV. RESULTS AND DISCUSSION

In this section, we will emphasize on the results of our ML models in this project.

A. Performance Comparison

Among the models we evaluated, ResNet50 achieved the highest test accuracy of about 98.22%. Among rest of the models,Random Forest and KNN gave similar accuracies, both above 65% and CNN gave higher accuracy of 81.5%. Comparatively, AdaBoost gave the lowest test accuracy of 24.11%.

TABLE I
PERFORMANCE COMPARISON OF ALL MODELS

Model	Test(%)	Train(%)	Validation(%)
Custom CNN	81.5	92.09	80.78
ResNet50	98.22	99.26	97.88
Random Forest	66.82	100.00	67.43
AdaBoost	24.11	24.31	24.31
KNN (k=5)	67.52	78.14	67.86



Fig. 6. Train and Validation accuracy comparison

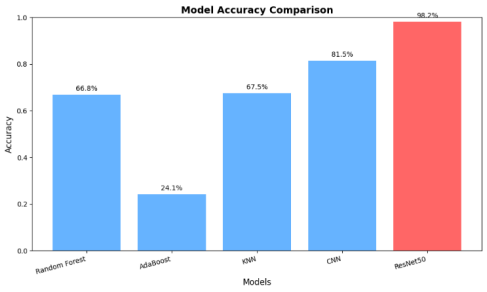


Fig. 7. Model Test Accuracy Comparison

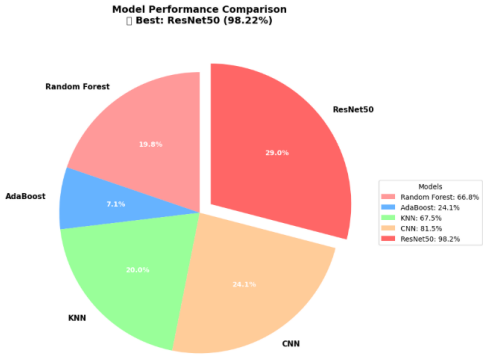


Fig. 8. Model Performance Comparison

B. Performance Matrix

We used a few machine learning models to detect plant disease from the given PlantVillage dataset. We started with the performance evaluation of five different models: CNN, Random Forest, Adaboost, KNN and Resnet50. Each model was examined according to its precision, recall, and F1 score.

In general, Precision is the ratio between the True Positives and all the Positives while Recall checks the number of true positive cases that the model can correctly identify [14]. Additionally, the F1 score is a metric used for imbalance classes in a data set that combines precision and recall into a single value [15].

The Custom CNN achieved the highest performance across all metrics with a recall of 91.02%, precision 91/11% and F1-Score of 90.92%. This indicates the custom CNN can generalize effectively unseen plant disease images. Although ResNet50 is a deeper learning model pretrained on ImageNet, fine tuning might improve its performance. Adaboost recorded the lowest performance with an F1 score of 19.33% . Rest of the two models Random Forest and KNN demonstrated moderate performance.

TABLE II
PERFORMANCE TABLE

Model	Precision(%)	Recall(%)	F1-Score(%)
Custom CNN	91.11	91.02	90.92
ResNet50	29.51	39.41	29.82
Random Forest	65.71	66.86	63.45
AdaBoost	20.01	25.55	19.33
KNN (k=5)	62.41	54.03	51.63

C. Confusion Matrix

We use confusion matrix to determine how well a classification model is performing. It shows where the model went right or wrong by comparing the model predictions with the actual result [16]. The confusion matrix gives us the opportunity to improve the model by learning where the model is making mistakes.

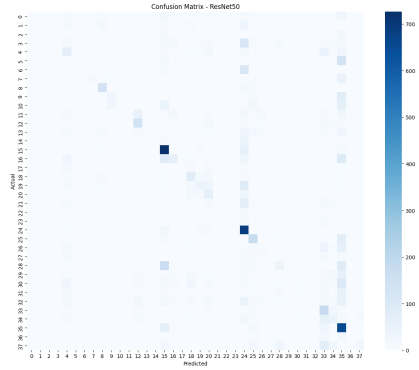


Fig. 9. Confusion Matrix

The confusion matrix for ResNet50 demonstrates significant class imbalance bias. It shows the strong prediction capability for classes 15, 24 and 35. Off-diagonals activation denote the samples belonging to underpresented classes are misclassified to dominant classes.

V. CONCLUSION

Despite some limitations, our models managed to perform well to detect plant disease under diverse environmental conditions. We implemented a Custom CNN, Random Forest, Adaboost, KNN and Resnet50 under a consistent data preprocessing. Our results conclude that deep learning models specially models with pretrained weights gain higher performance for image datasets like the PlantVillage dataset used in our project. Although, classical ML models are computationally lighter, they become constrained by their feature representation quality when applied to high-dimensional image data.Our study highlights the trade-offs between model perfrmance.With the help of further research in the future, we can build better models to detect plant disease faster and contribute to the wellbeing of our society.

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